

Color Image Processing

Pengolahan Citra Digital



- 
- 
- For a long time I limited myself to one color – as a form of discipline.

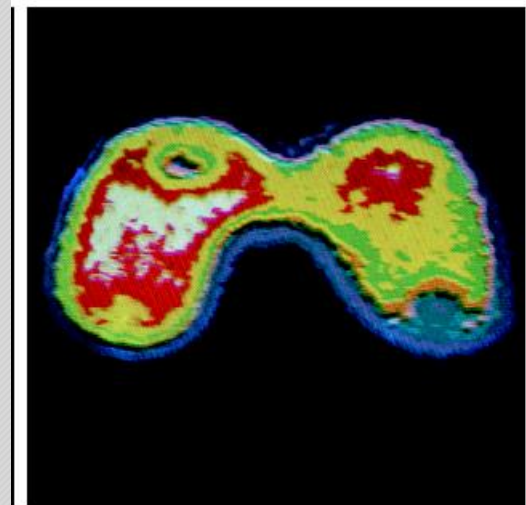
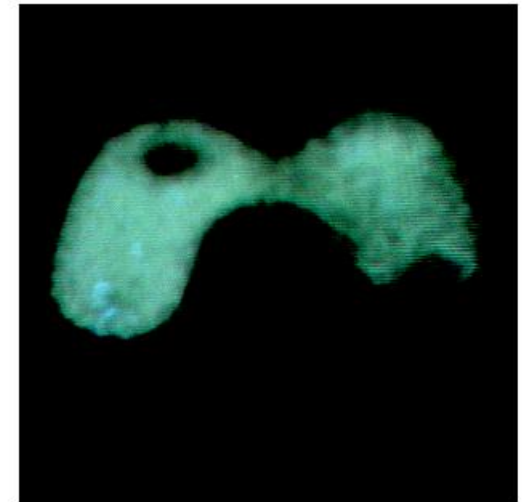
- Pablo Picasso

- It is only after years of preparation that the young artist should touch color – not color used descriptively, that is, but as a means of personal expression.

- Henri Matisse

Preview

- Why use color in image processing?
 - Color is a **powerful descriptor**
 - Object identification and extraction
 - eg. Face detection using skin colors
 - Humans can **discern** thousands of color shades and intensities
 - c.f. Human discern only two dozen shades of grays



Preview (cont.)

- Two category of color image processing
 - Full color processing
 - Images are acquired from full-color sensor or equipments
 - Pseudo-color processing
 - In the past decade, color sensors and processing hardware are not available
 - Colors are assigned to a range of monochrome intensities

OUTLINE

Color fundamentals

Color models

Basics of full-color image processing

Color transformations

Smoothing and sharpening

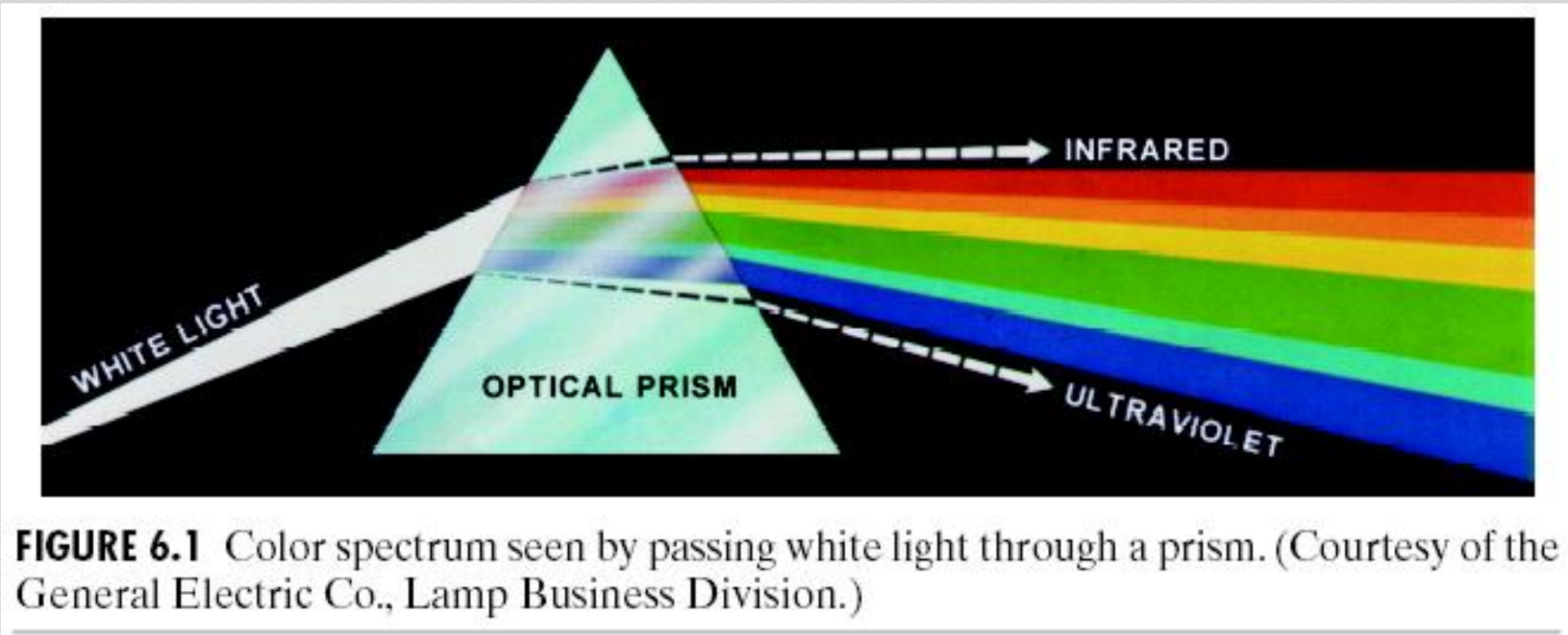


Color fundamentals

- **Physical** phenomenon
 - Sifat fisik warna diketahui
- **Physio-psychological** (fisio-psikologis) phenomenon
 - Bagaimana otak manusia memahami dan menafsirkan warna?

Color fundamentals (cont.)

- 1666, Isaac Newton



Warna Tampak

- Warna sinar yang dapat terlihat mata adalah visible spectrum (sinar tampak) dengan Panjang gelombang dari 400 sampai 700 nm

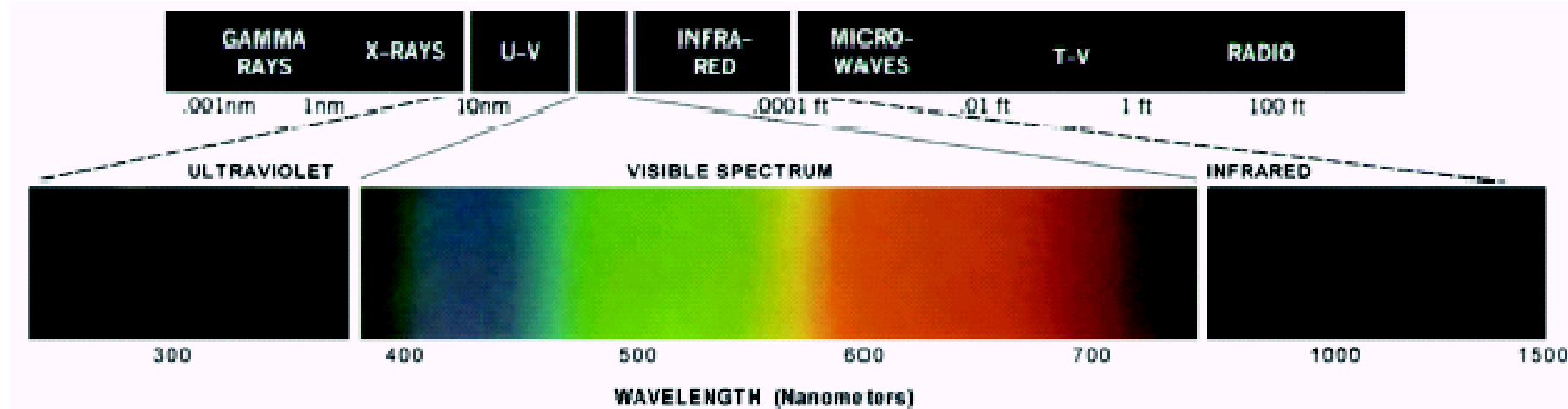
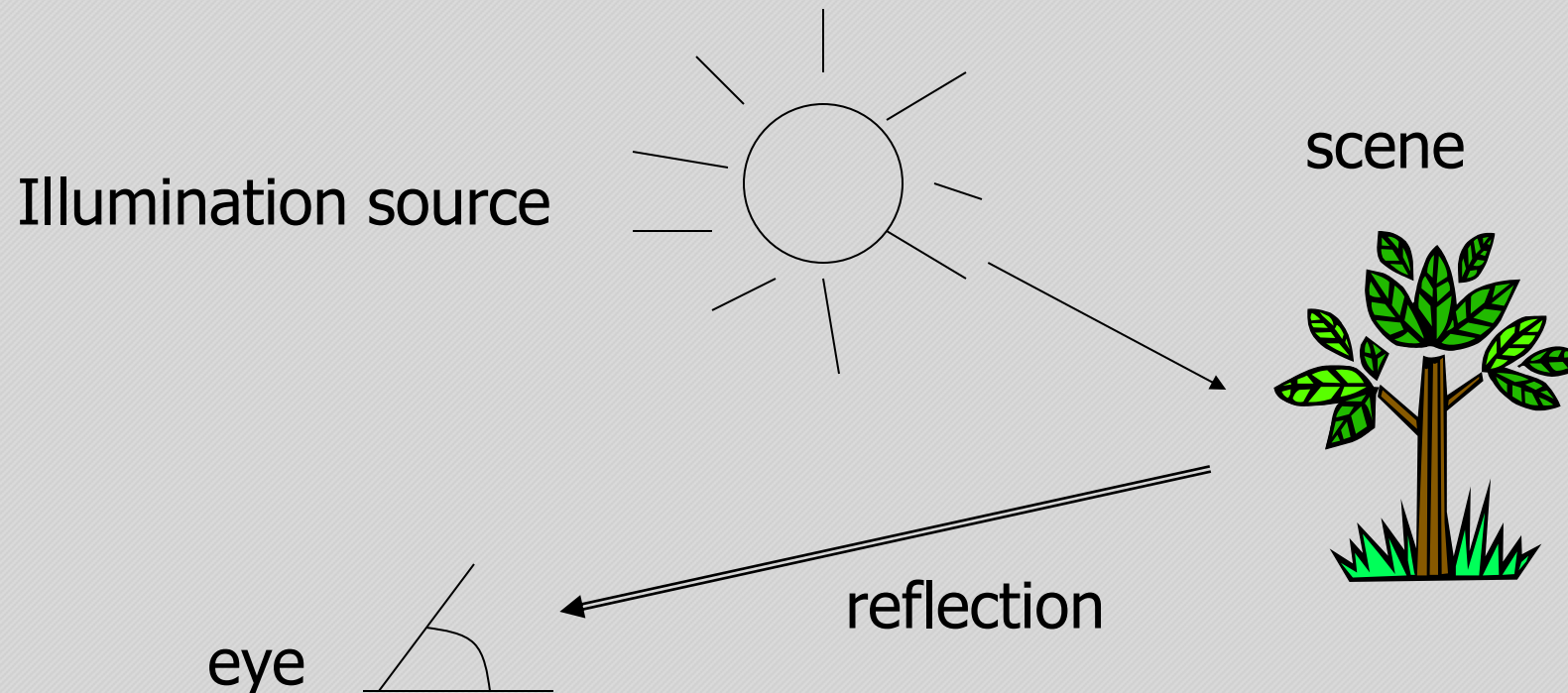


FIGURE 6.2 Wavelengths comprising the visible range of the electromagnetic spectrum. (Courtesy of the General Electric Co., Lamp Business Division.)

Color fundamentals (cont.)

- Warna yang dirasakan manusia pada suatu objek = cahaya yang dipantulkan dari objek

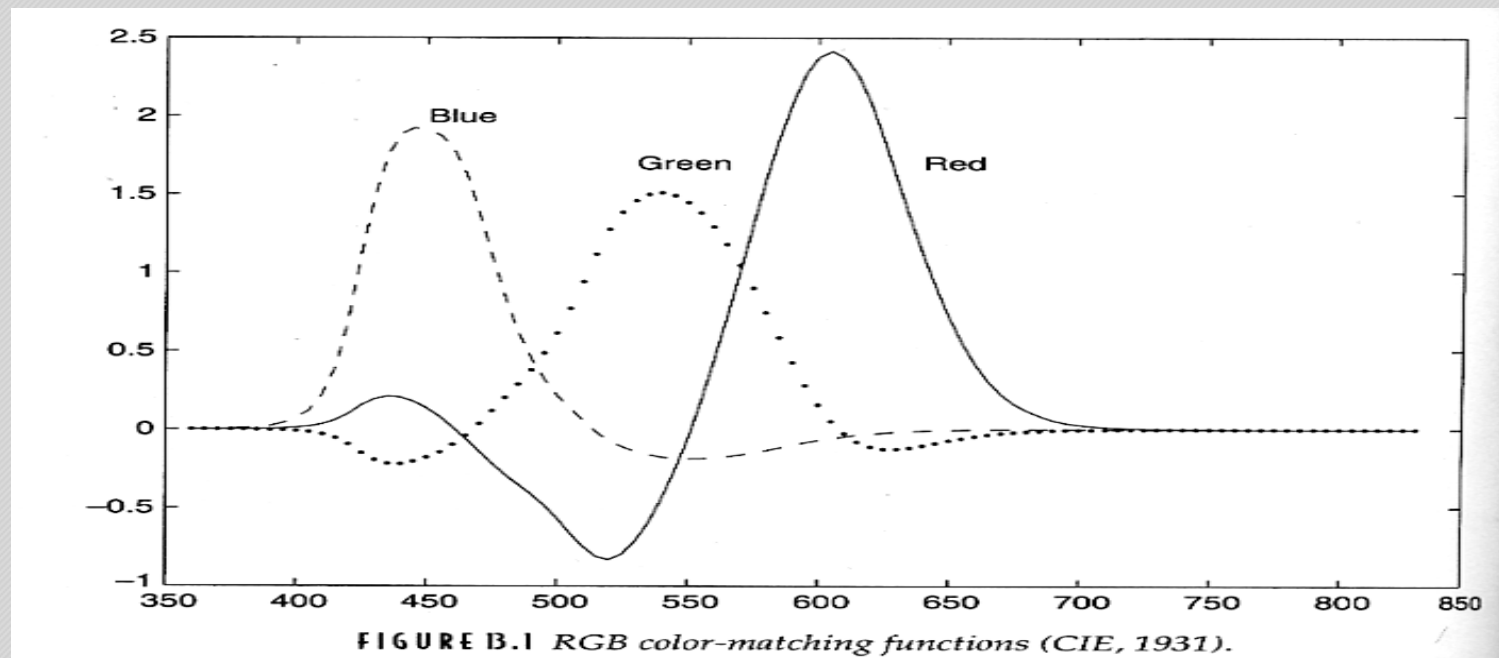


chromatic light: (panjang gelombang: 400~700nm)

- Tiga besaran pokok digunakan untuk menyatakan kualitas dari sumber cahaya berwarna:
- radiance (pancaran): jumlah total energi yang mengalir dari sumber cahaya (dalam watt)
- luminance: jumlah energi yang dirasakan pengamat dari sumber cahaya (dalam lumen, lm); misalnya, inframerah jauh cahaya memiliki pancaran yang besar, tetapi pengamat tidak akan melakukannya rasakan itu
- Brightness (kecerahan): (deskriptor subjektif) mirip dengan gagasan akromatik tentang intensitas, salah satu faktor kunci dalam menggambarkan sensasi warna

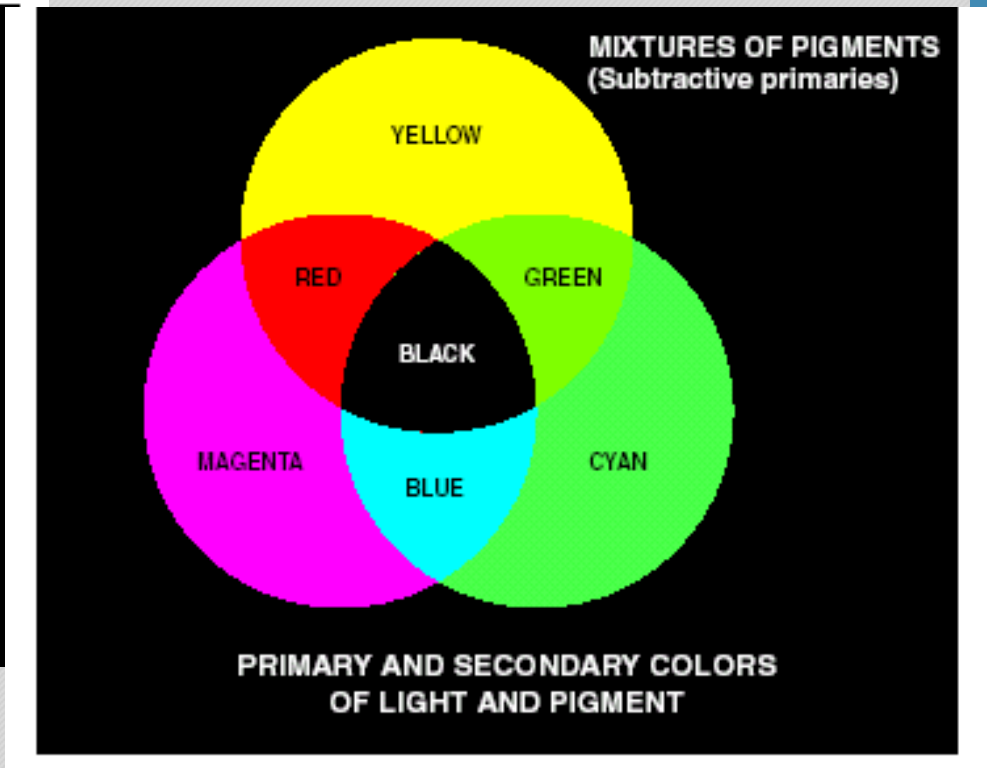
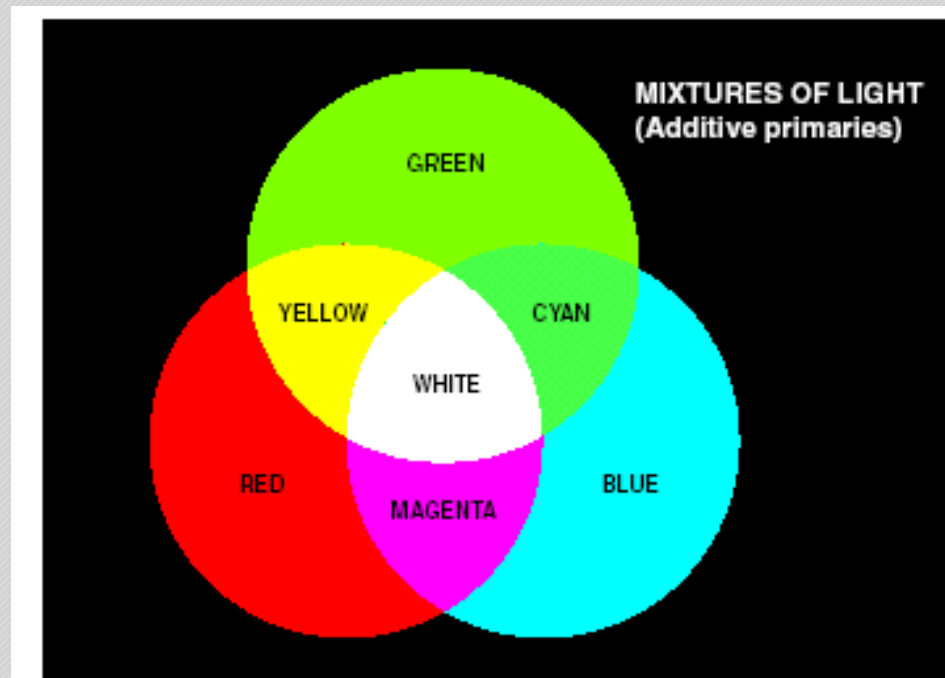
How human eyes sense light?

- 6~7M Cones are the sensors in the eye
- 3 principal sensing categories in eyes
 - Red light 65%, green light 33%, and blue light 2%



Primary and secondary colors

- In 1931, CIE(International Commission on Illumination) defines specific wavelength values to the primary colors
 - B = 435.8 nm, G = 546.1 nm, R = 700 nm
 - However, we know that no single color may be called red, green, or blue
- Secondary colors: G+B=Cyan, R+G=Yellow, R+B=Magenta

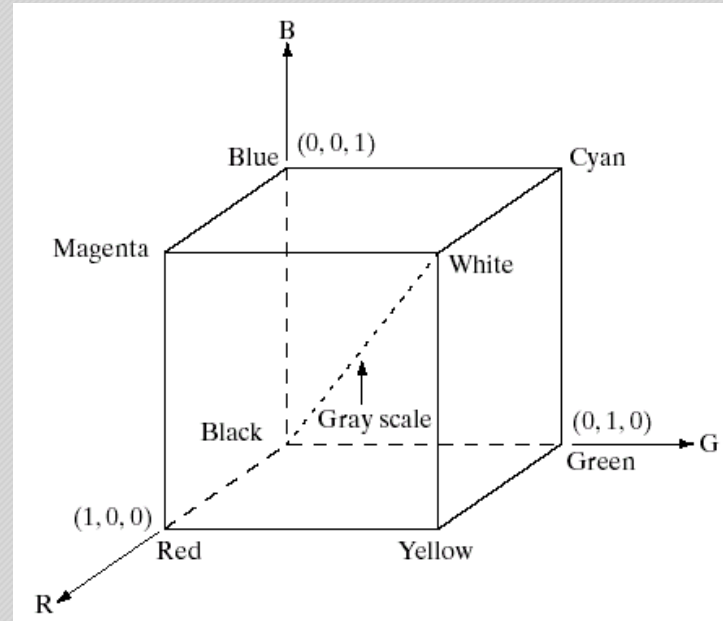


Color models (MODEL WARNA)

- Model warna adalah metode untuk menentukan warna menurut suatu standar tertentu. Secara umum biasanya terdiri atas sistem koordinat tiga dimensi dan dalam setiap *subspace*-nya warna diwakili oleh satu titik tunggal.
- Model warna yang digunakan di dalam pengolahan citra yaitu:
 1. Model warna RGB
 2. Model warna CMY dan CMYK
 3. Model warna YCbCr
 4. Model warna HSI → sesuai deskripsi manusia
 5. Model warna

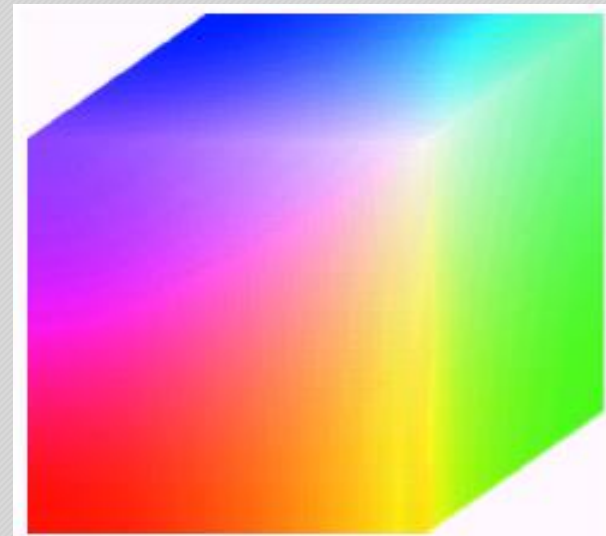
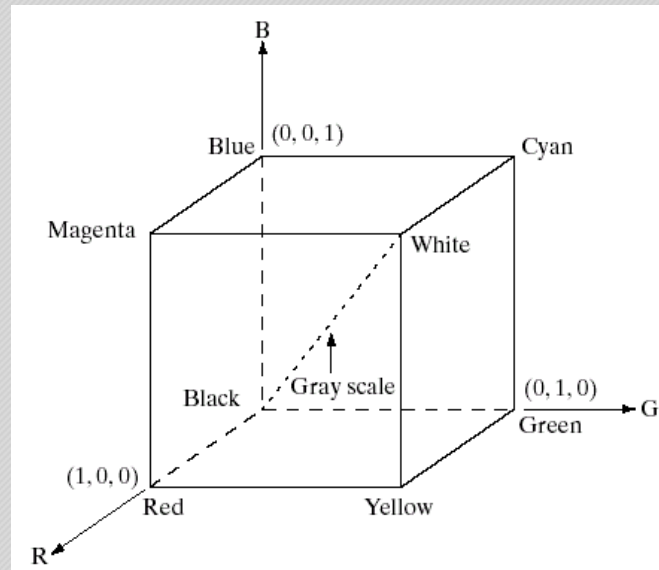
RGB color model

- Setiap warna yang tampak merupakan kombinasi dari warna utama yaitu red (R), green (G) dan blue (B)
- Model ini didasarkan pada system koordinat kartesian
- Model warna RGB digunakan untuk display pada layer komputer



Pixel depth

- **Pixel depth**: jumlah **bits** yang digunakan untuk merepresentasikan setiap piksel di ruang RGB
- **Citra Full-color** : 24-bit citra warna RGB
 - $(R, G, B) = (8 \text{ bits}, 8 \text{ bits}, 8 \text{ bits})$





Full color

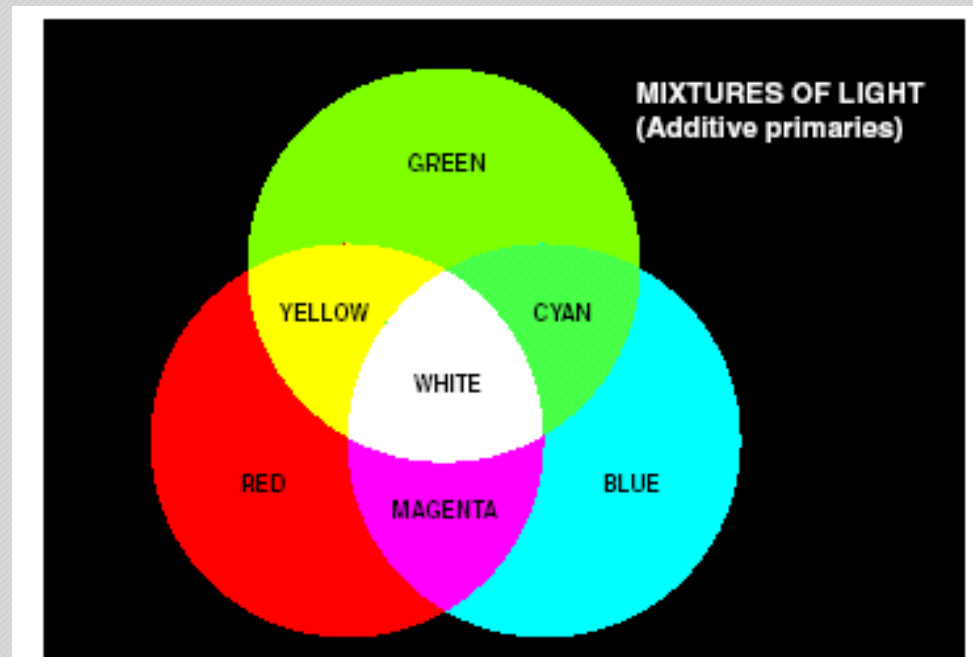


Citra warna dan komponen RGB

CMY model (+Black = CMYK)

- **CMY**: warna sekunder dari sinar, atau warnakomplemen dari warna primar
- Digunakan untuk printer yang mencetak hardcopy citra warna

$$\begin{bmatrix} C \\ M \\ Y \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} - \begin{bmatrix} R \\ G \\ B \end{bmatrix}$$

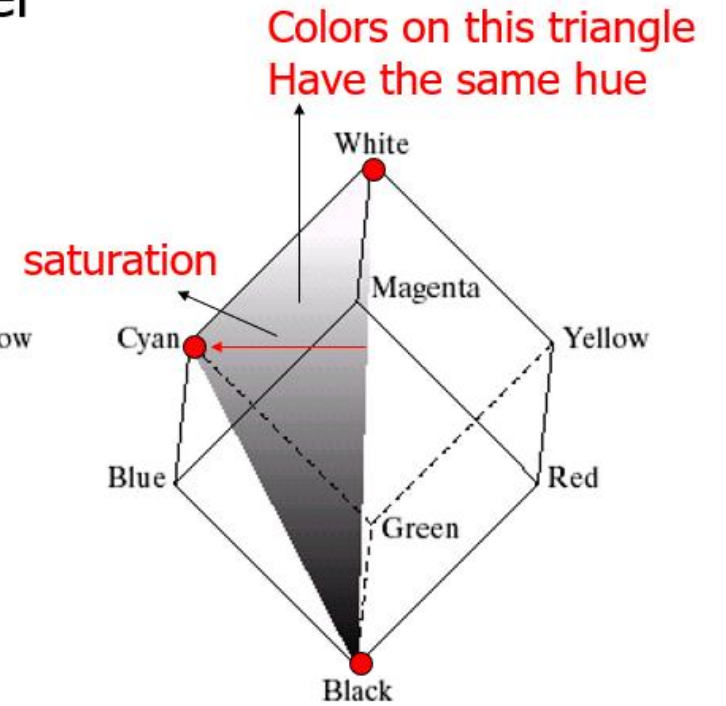
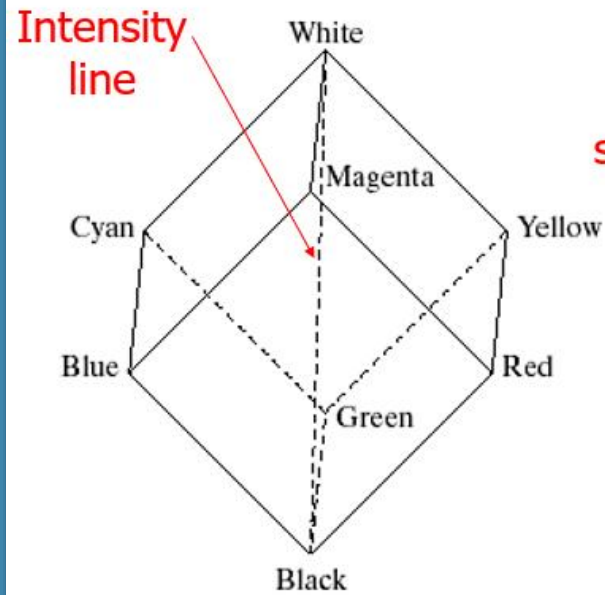


HSI color model

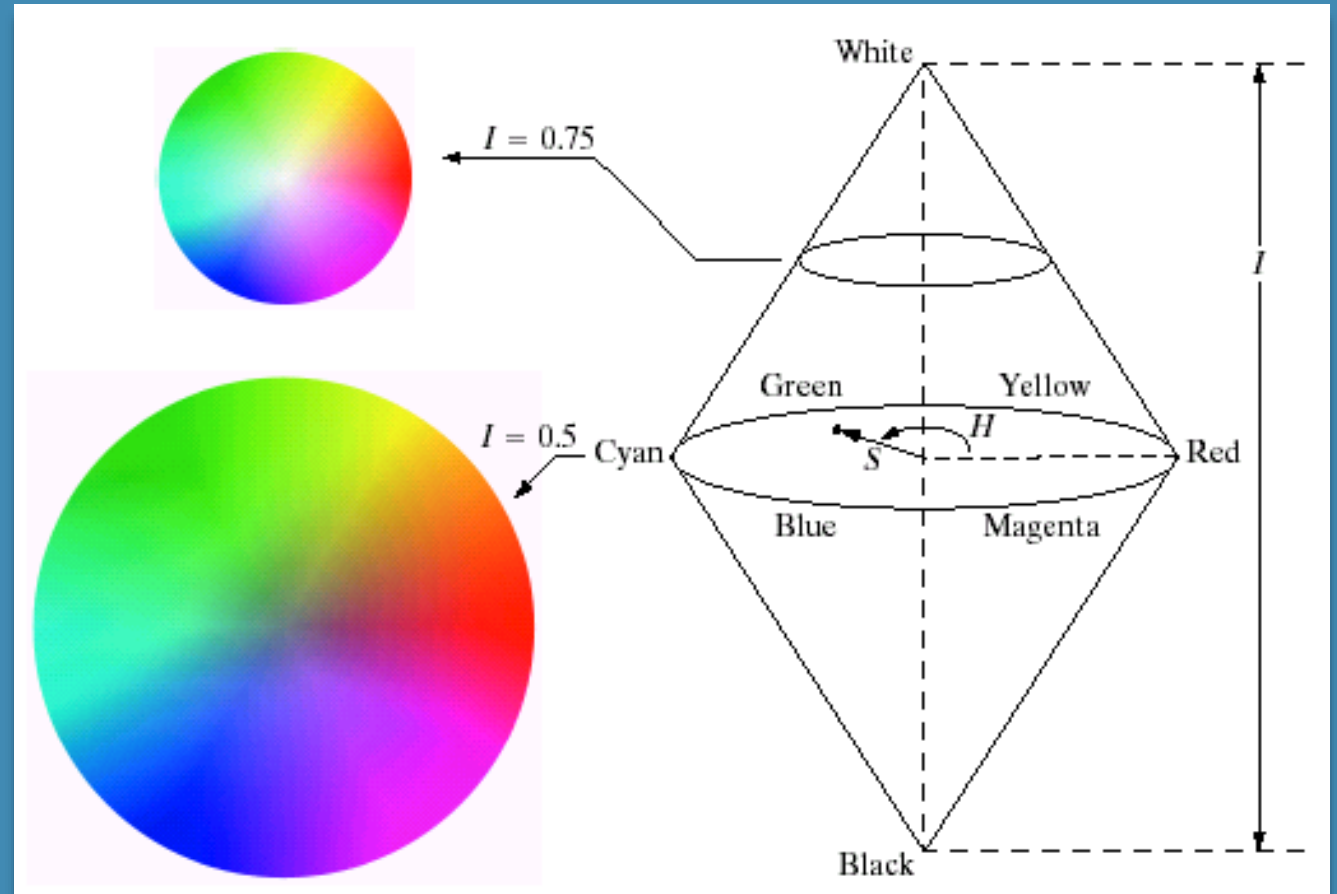
- Manusia tidak lazim mendeskripsikan warna dalam komponen R, G dan B. Misalkan menyatakan warna coklat dalam prosentase R, G dan B
- Manusia mendeskripsikan warna dengan atribut hue(H), saturation(S), and Intensitas (I) (brightness)
 - Hue 色度: color attribute
 - Saturation: purity of color (white->0, primary color->1)
 - Intensity (Brightness): achromatic notion

HSI color model (cont.)

■ RGB -> HSI model

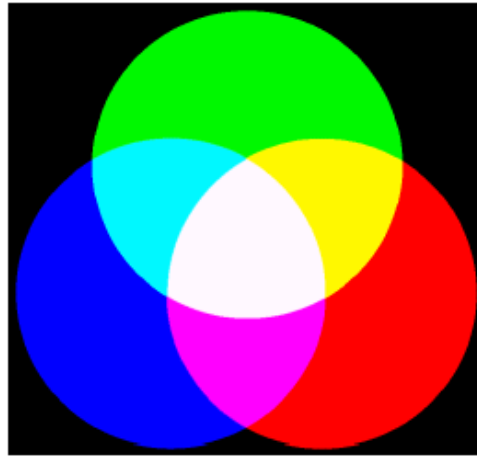


HSI model

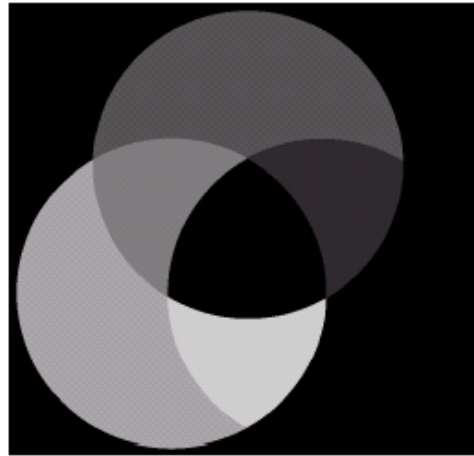


HSI component images

R,G,B



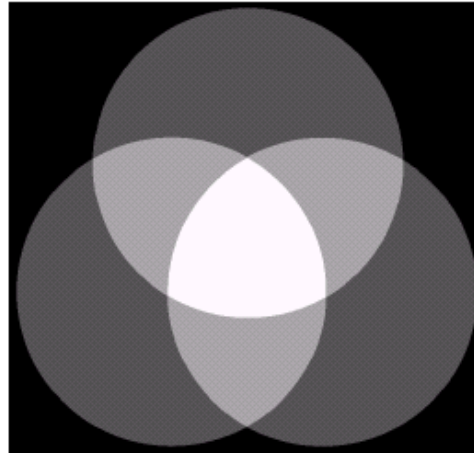
Hue



saturation



intensity



Konversi RGB → HSI

$$H = \begin{cases} \theta, & B \leq G \\ 360 - \theta, & B > G \end{cases} \text{ where}$$

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R - G) + (R - B)]}{\sqrt{(R - G)^2 + (R - B)(G - B)}} \right\}$$

$$S = 1 - \frac{3}{R + G + B} [\min(R, G, B)]$$

$$I = \frac{1}{3}(R + G + B)$$

Konversi HIS → RGB

$$B = I(1 - S)$$

$$R = I \left[1 + \frac{S \cos H}{\cos(60^\circ - H)} \right]$$

$$G = 1 - (R + B)$$

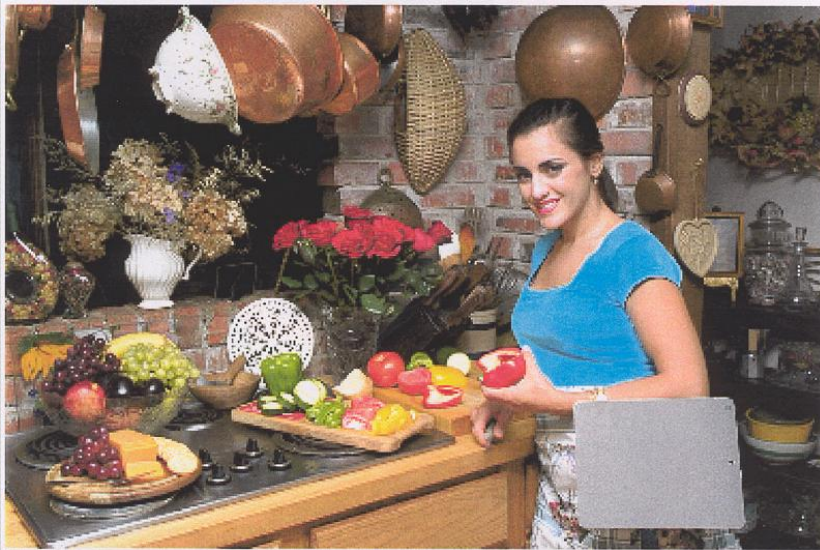
Problem pada Pemrosesan Citra Warna

- Saat memproses gambar berwarna, masalah berikut (antara lain) harus ditangani:
 - Gambar adalah vektor $\rightarrow$ 3 angka terkait dengan setiap piksel
 - Warna yang direkam oleh kamera sangat bergantung pada kondisi pencahayaan



Kondisi Pencahayaan

- Kondisi pencahayaan pemandangan memiliki pengaruh besar pada warna yang direkam.



Gambar yang diambil diterangi oleh flash.



Gambar diambil diterangi oleh lampu pijar

Empat gambar berikut dari pemandangan yang sama diperoleh dalam kondisi pencahayaan yang berbeda

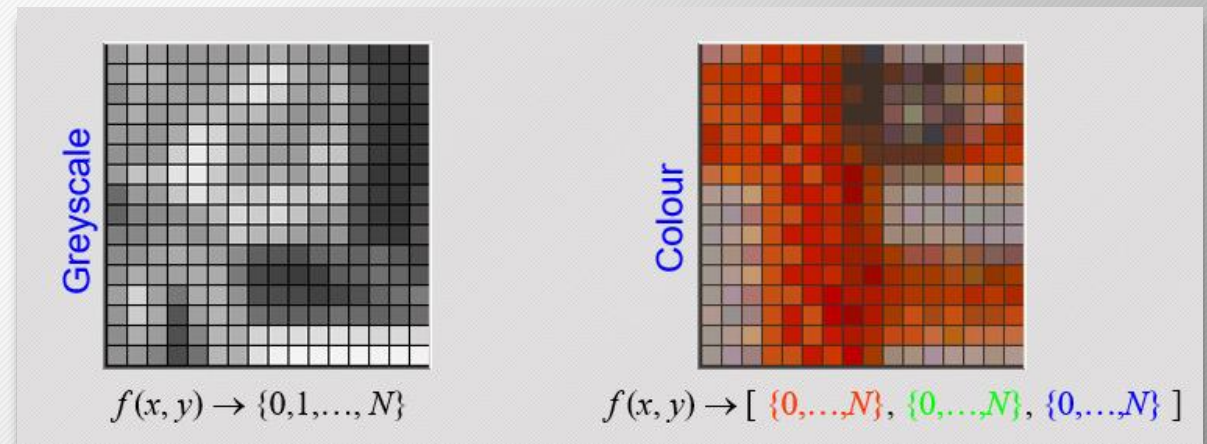


Menangani Perubahan Pencahayaan

- Mengetahui nilai RGB saja tidak cukup untuk mengetahui segala sesuatu tentang gambar.
 - The R, G and B yang digunakan oleh perangkat yang berbeda biasanya berbeda.
- Untuk karya ilmiah, kamera dan pencahayaan harus dikalibrasi.
- Untuk aplikasi multimedia, ini lebih sulit diatur:
 - Ada algoritma untuk memperkirakan warna iluminasi.

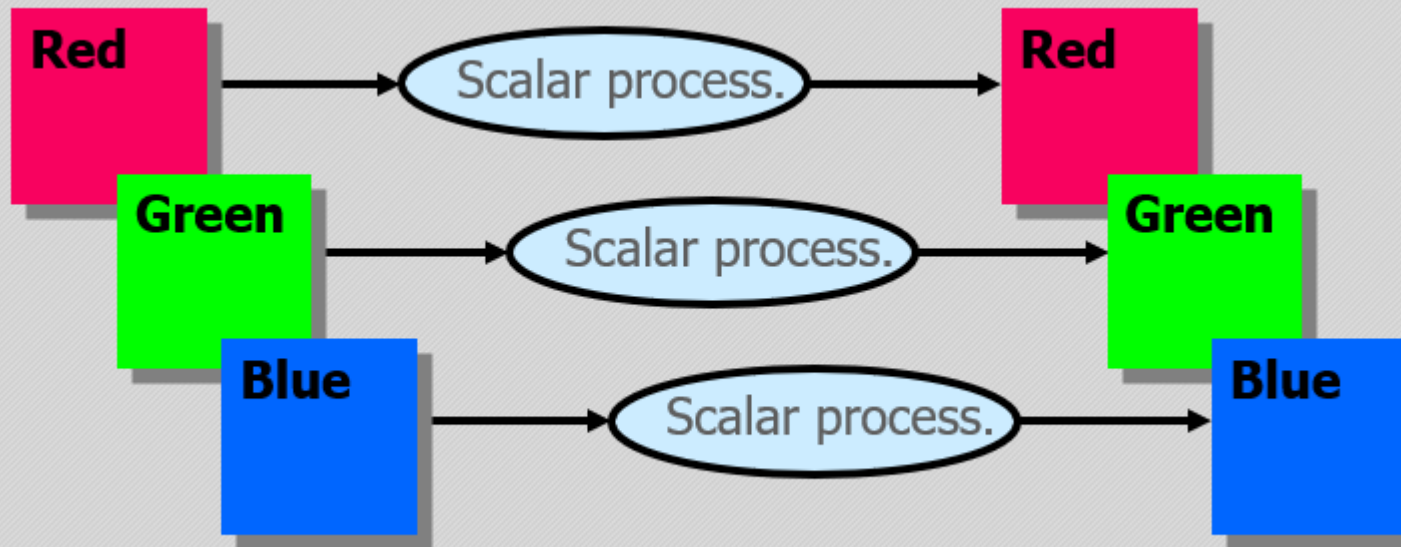
Pemrosesan citra vector (warna)

- Sebuah citra warna memiliki vektor pada setiap piksel. Untuk gambar warna, vektor-vektor ini masing-masing memiliki 3 komponen.
- Citra vektor dengan jumlah komponen yang lebih banyak juga ada, mis. dalam citra satelit.
- Ada dua cara untuk memproses citra vektor:
 - Pemrosesan marjinal.
 - Pemrosesan vektor.



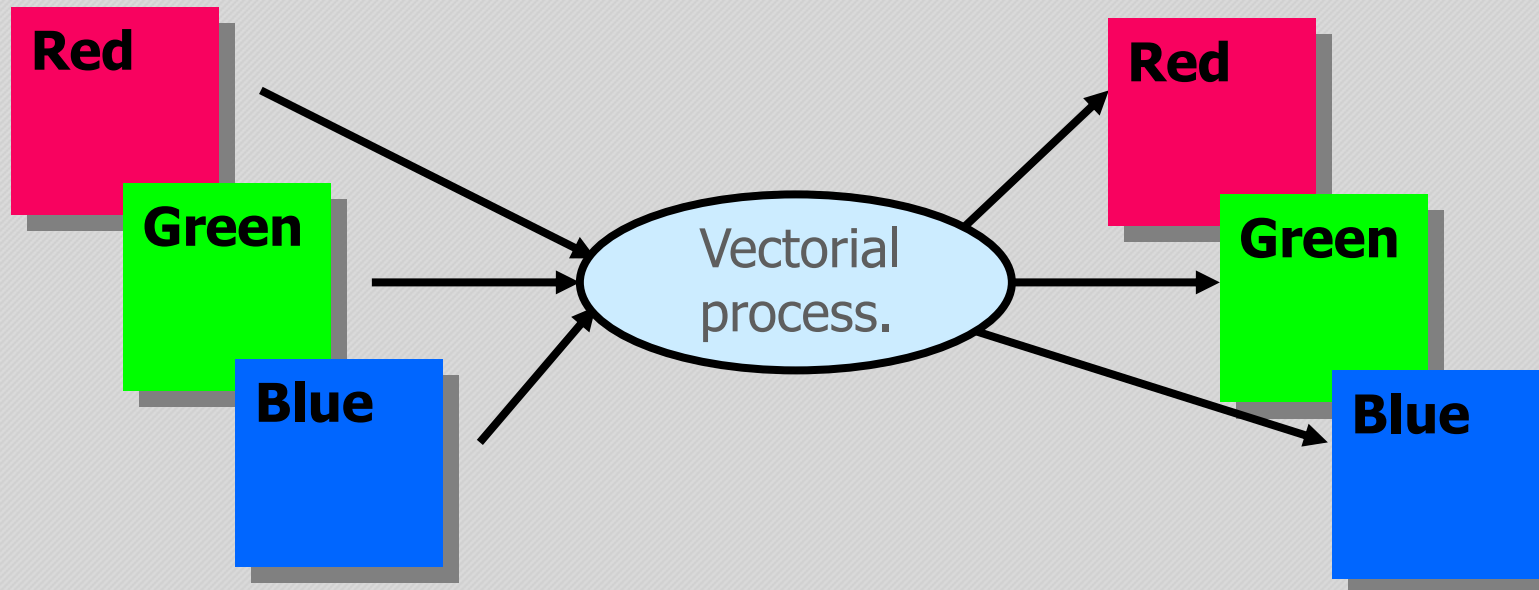
Pemrosesan Marjinal

- Setiap chanel diproses secara terpisah:



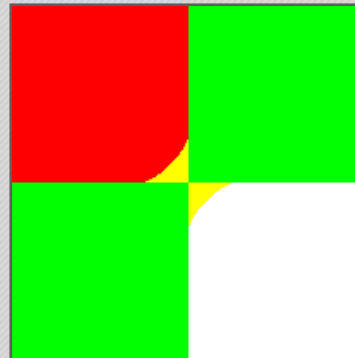
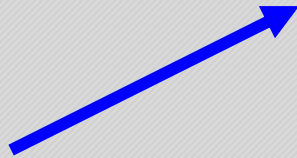
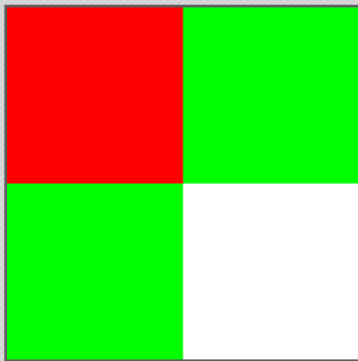
Vectorial Processing

- Ke-tiga warna diproses sebagai unit tunggal :

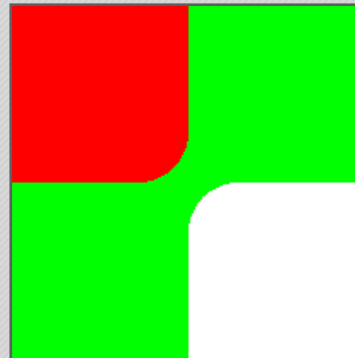
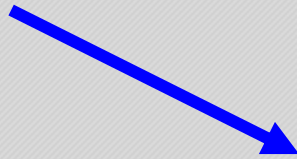


Masalah Warna Palsu

False
Colours !!

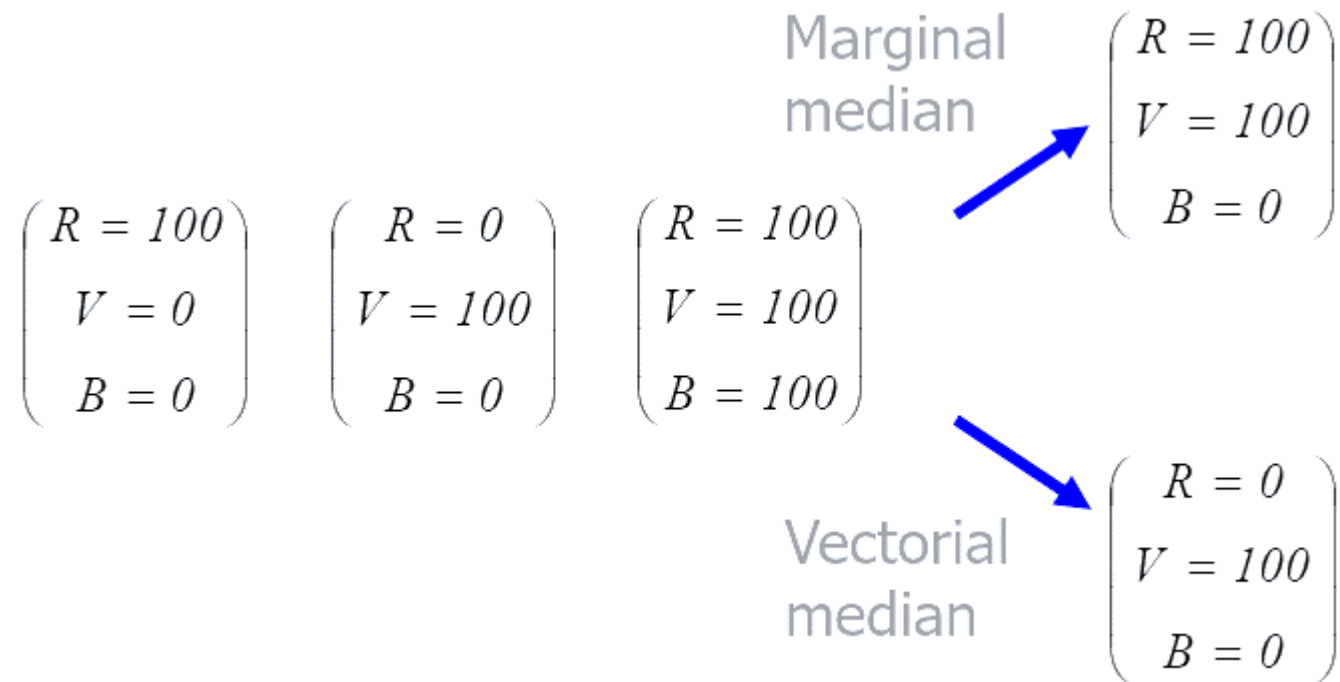


Marginal
Median



Vectorial
Median

Masalah Warna Palsu



Color transformation

- Transformation within a single color model

$$g(x, y) = T[f(x, y)]$$

$$s_i = T_i(r_1, r_2, \dots, r_n), i = 1, \dots, n$$

- Examples:

$$g(x, y) = kf(x, y), 1 < k < 1$$

$$HSI : s_3 = kr_3$$

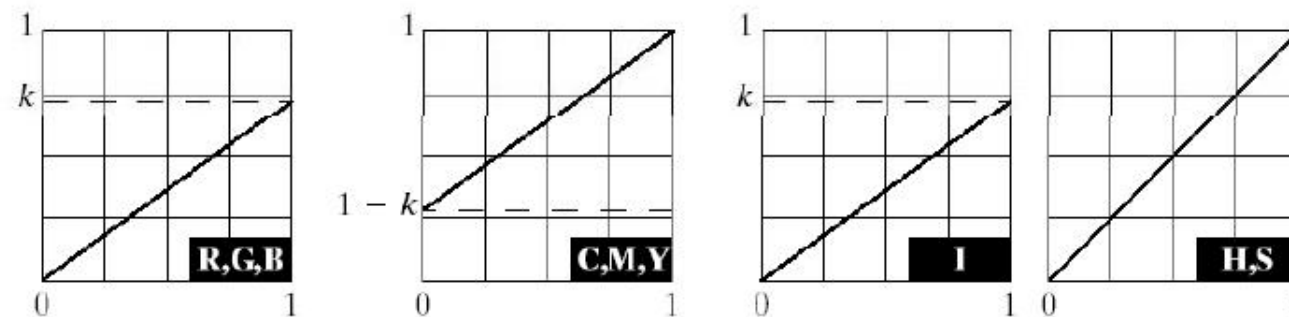
$$RGB : s_i = k_i r_i, i = 1, 2, 3$$

$$CMY(K) : s_i = kr_i + (1 - k), i = 1, 2, 3$$

a b
c d e

FIGURE 6.31

Adjusting the intensity of an image using color transformations. (a) Original image. (b) Result of decreasing its intensity by 30% (i.e., letting $k = 0.7$). (c)–(e) The required RGB, CMY, and HSI transformation functions. (Original image courtesy of MedData Interactive.)



Color Complements

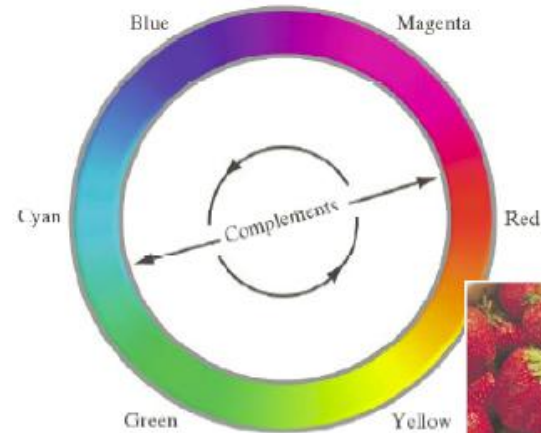
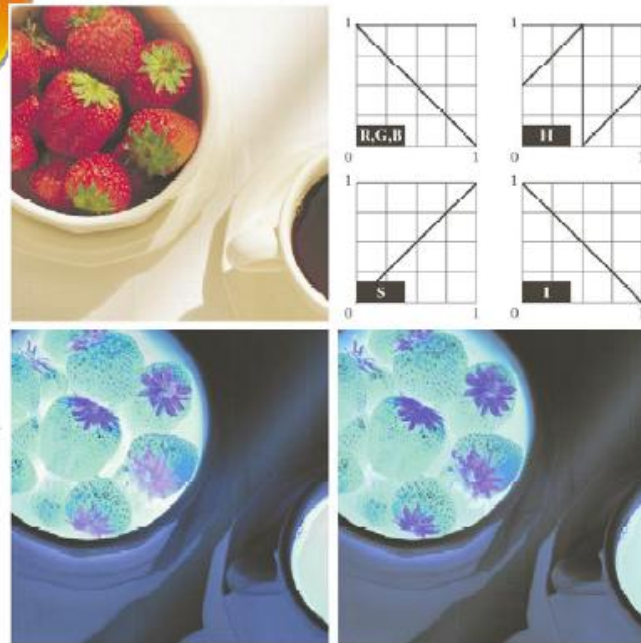


FIGURE 6.32
Complements on
the color circle.



a	b
c	d

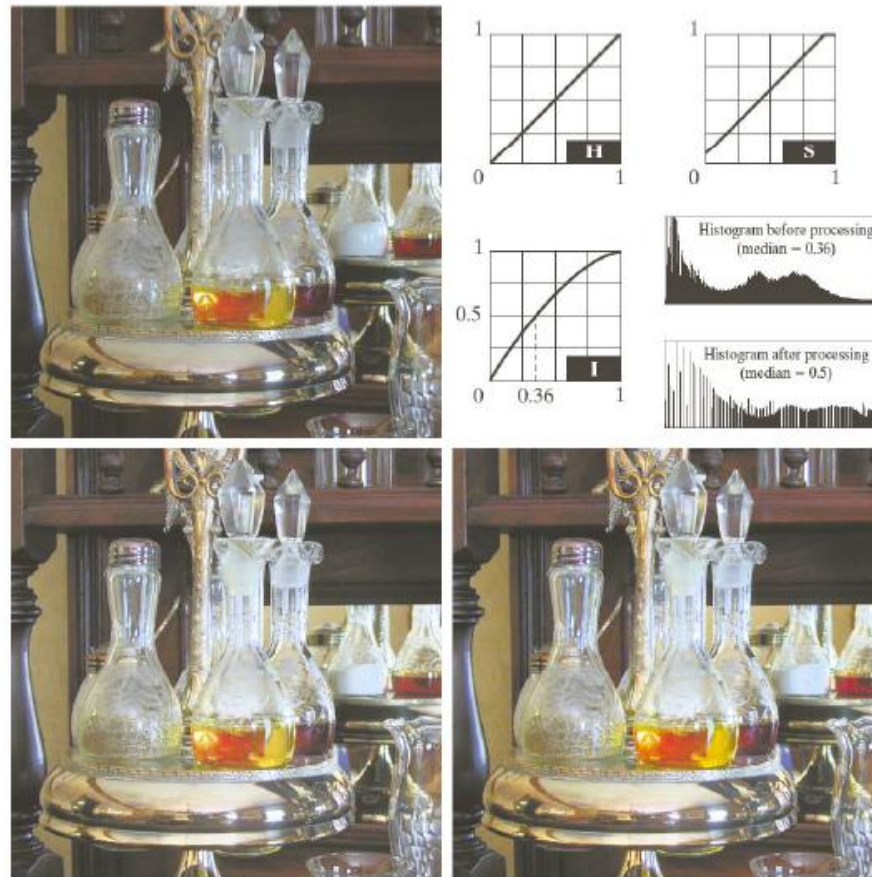
FIGURE 6.33

Color
complement
transformations.
(a) Original
image.

(b) Complement
transformation
functions.

(c) Complement
of (a) based on
the RGB mapping
functions. (d) An
approximation
of the RGB
complement
using HSI
transformations.

Histogram Processing

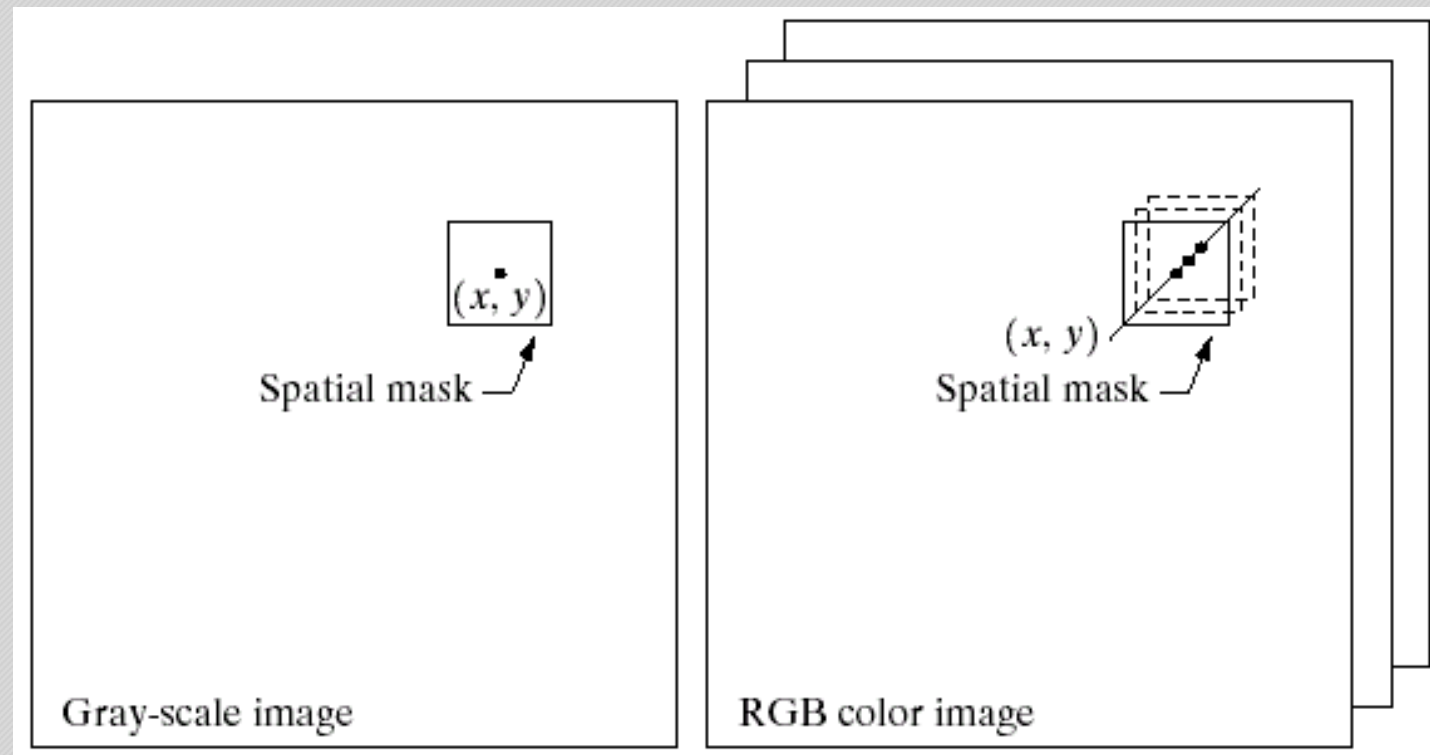


a b
c d

FIGURE 6.37
Histogram
equalization
(followed by
saturation
adjustment) in the
HSI color space.

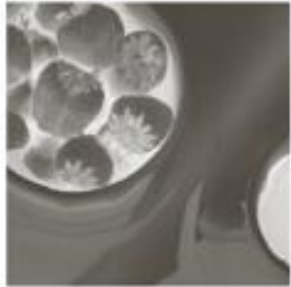
Pelembutan Citra Warna

- Neighborhood processing





Full color



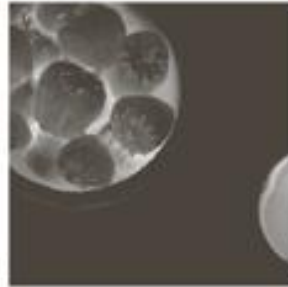
Cyan



Magenta



Yellow



Black



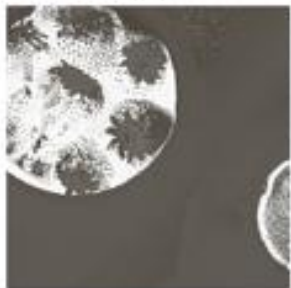
Red



Green



Blue



Hue



Saturation



Intensity

FIGURE 6.30 A full-color image and its various color-space components. (Interactive.)

Pelembutan Citra Warna : Penapis Rerata

$$\bar{\mathbf{c}}(x, y) = \frac{1}{K} \sum_{(x, y) \in S_{xy}} \mathbf{c}(x, y)$$

vector processing

Neighborhood
Centered at (x,y)

$$\bar{\mathbf{c}}(x, y) = \begin{bmatrix} \frac{1}{K} \sum_{(x, y) \in S_{xy}} R(x, y) \\ \frac{1}{K} \sum_{(x, y) \in S_{xy}} G(x, y) \\ \frac{1}{K} \sum_{(x, y) \in S_{xy}} B(x, y) \end{bmatrix}$$

per-component processing

Sumber: Jen-Chang Liu, Spring 2006
Color Image Processing

original



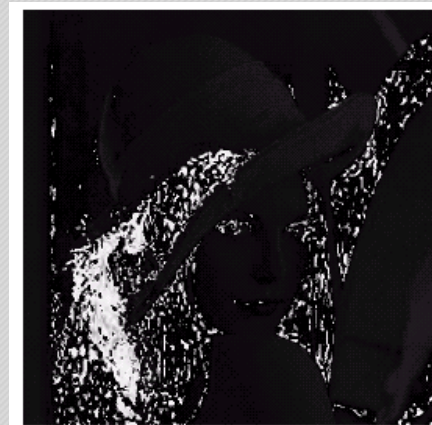
R



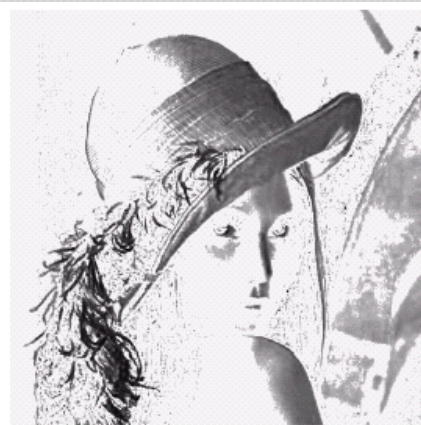
G



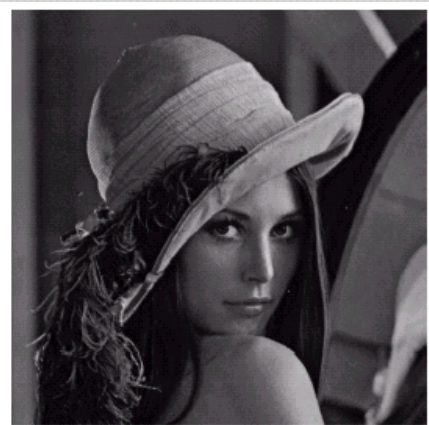
B



H



S

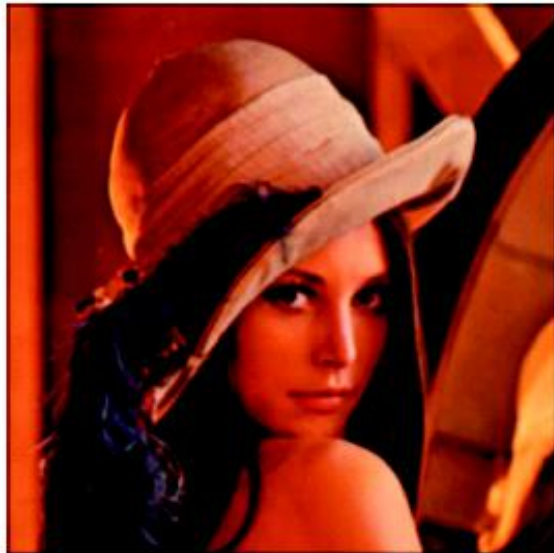


I

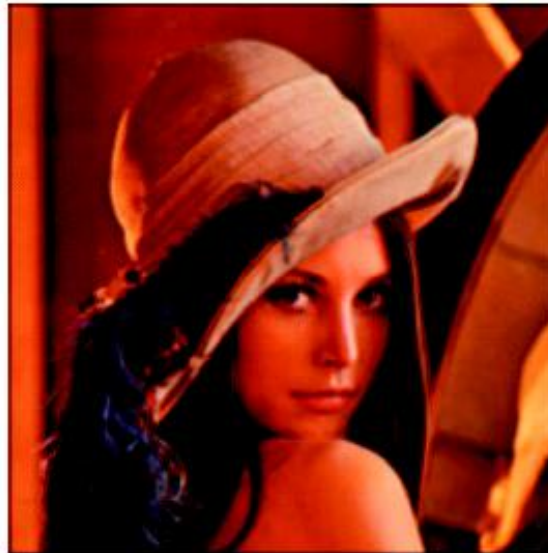


Example: 5x5 smoothing mask

RGB model



Smooth I
in HSI model



difference



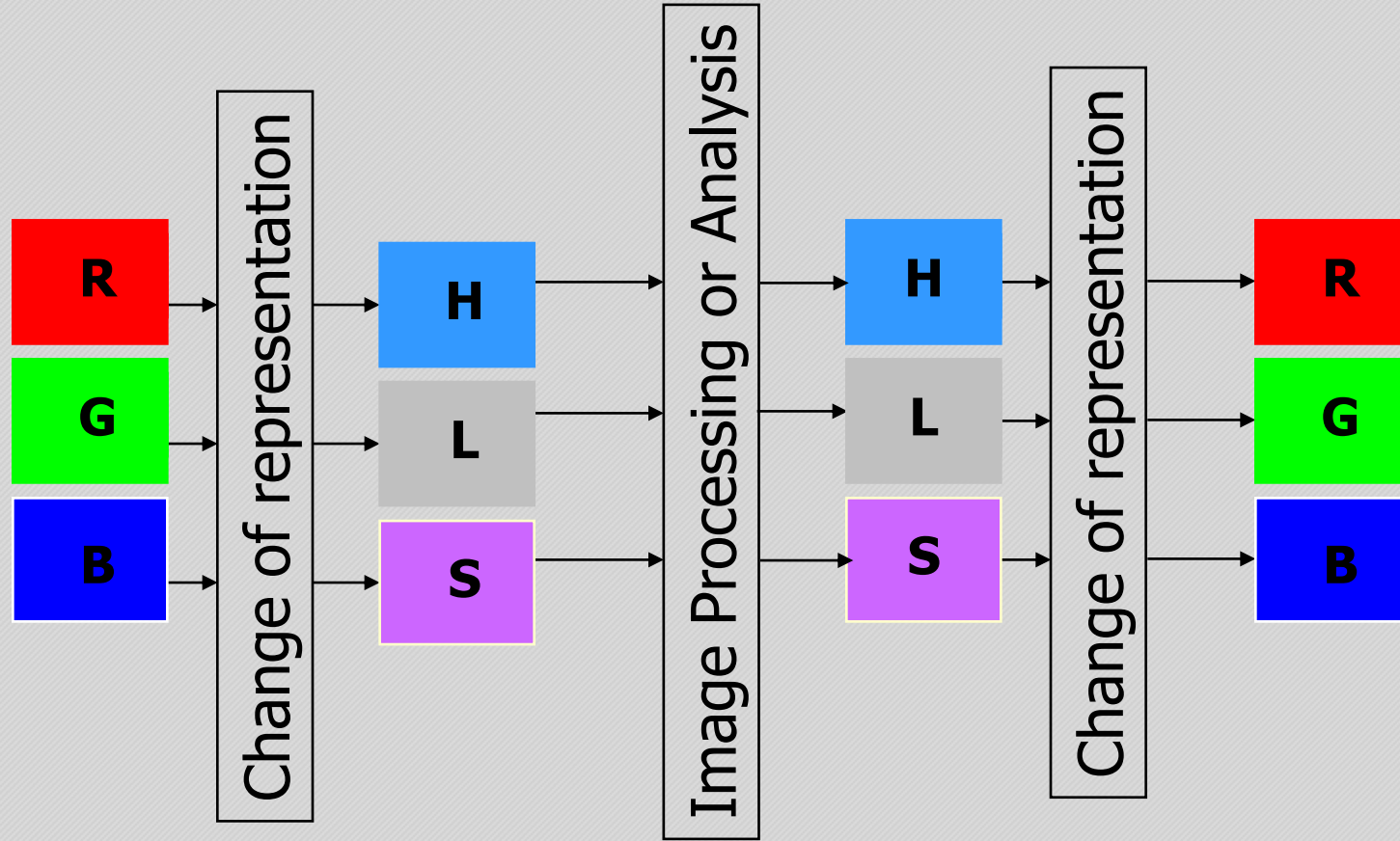
a b c

FIGURE 6.40 Image smoothing with a 5×5 averaging mask. (a) Result of processing each RGB component image. (b) Result of processing the intensity component of the HSI image and converting to RGB. (c) Difference between the two results.

Processing and Analysing Colour Images

- Some applications which take advantage of the good properties of the IHLS space are presented.
- The following are discussed:
 - Hue statistics.
 - Mathematical morphology on colour images.

Basic Idea



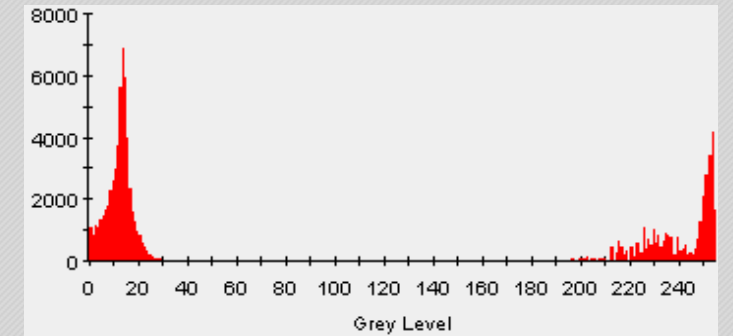
Processing the Hue Component



Colour image – “The virgin”,
P. Serra (16th century)



Hue component



Histogram of the hue

The reds and violets are separated by a large discontinuity, even though they look similar.

Statistika Hue

- Untuk kecerahan dan saturasi, seseorang dapat menggunakan metode statistik standar untuk menghitung rata-rata, simpangan baku, dll.
- Untuk data a_i ($i = 1, 2, \dots, n$) yang terdistribusi pada lingkaran satuan, arah rata-rata adalah vektor resultan yang diperoleh dengan menjumlahkan vektor satuan dengan arah a_i .
- Ukuran variansi dalam arah data diberikan oleh panjang vektor dibagi dengan n (panjang rata-rata), yang memiliki karakteristik sebagai berikut:
 - rentang $[0:1]$
 - Nilai mendekati 1 $\Rightarrow$ data kurang menyebar.

Mean

- Given n values of the hue H_i
- The mean direction $\overline{H}$ is calculated as follows:

$$A = \sum_{i=1}^n \cos H_i, \quad B = \sum_{i=1}^n \sin H_i, \quad R^2 = A^2 + B^2 \quad (1)$$

$$\overline{H} = \begin{cases} \arctan\left(\frac{B}{A}\right) & \text{if } B > 0, A > 0 \\ \arctan\left(\frac{B}{A}\right) + \pi & \text{if } A < 0 \\ \arctan\left(\frac{B}{A}\right) + 2\pi & \text{if } B < 0, A > 0 \end{cases} \quad (2)$$

- The mean length $\overline{R}$ is:

$$\overline{R} = \frac{R}{n} \quad (3)$$

- Let S_i be the saturation associated with hue H_i .

- We replace equation (1) by:

$$A_S = \sum_{i=1}^n S_i \cos H_i, \quad B_S = \sum_{i=1}^n S_i \sin H_i, \quad R_S^2 = A_S^2 + B_S^2$$

- To calculate the saturation-weighted mean direction H_S , we replace A and B by A_S and B_S in equation (2).
- Equation (3) becomes:

$$\bar{R}_S = \frac{R_S}{\sum_{i=1}^n S_i}$$

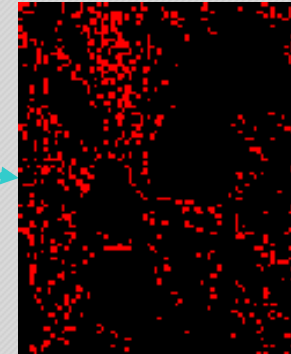
- Note that $\bar{R}_S$ remains a measure of the angular dispersion, and does not give information on the mean of the saturation.

Example

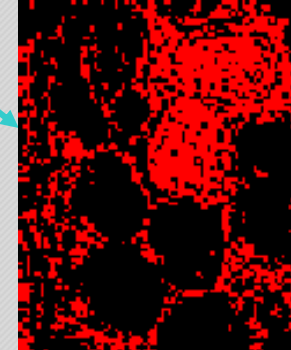
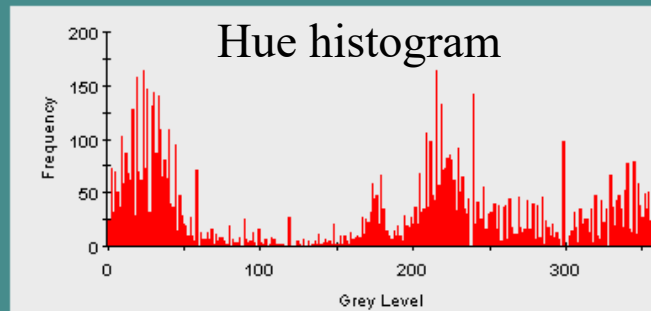
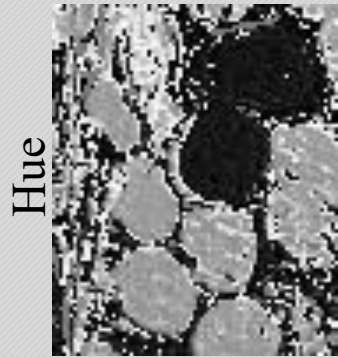


- $\bar{H} = 327^\circ$
- $\bar{R} = 0.24$

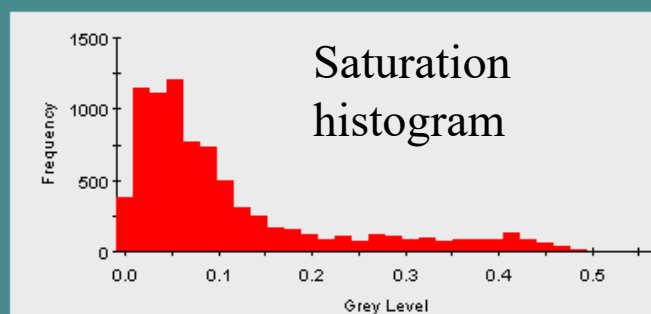
- Weighted
- $\bar{H}_S = 20^\circ$
 - $\bar{R}_S = 0.16$



Hue threshold
 $307^\circ - 347^\circ$



Hue threshold
 $0^\circ - 40^\circ$



Contoh Program

```
clear all;
clc;
I=imread('lena_color.jpg');
figure;imshow(I);

% proses marginal
R=I(:,:,1);
G=I(:,:,2);
B=I(:,:,3);

RH= histeq(I(:,:,1));
GH= histeq(I(:,:,2));
BH= histeq(I(:,:,3));

CitraHsl=cat(3,RH,GH,BH);
figure;imshow(CitraHsl);
```

```
%vektor hsv
Ihsv = rgb2hsv(I);
Ihsv(:,:,2) = histeq(Ihsv(:,:,2));
CitraHsl2 = hsv2rgb(Ihsv);
figure;imshow(CitraHsl2);

%vektor ntsc
J=rgb2ntsc(I);
J(:,:,1)=histeq(J(:,:,1));
JJ=ntsc2rgb(J);
imshow(I)
figure, imshow(JJ)
```

TUGAS

1. Lakukan pemfilteran dengan high pass filter pada citra warna dengan mengolah pada masing-masing lapisan warna (RGB) dan mengolah pada domain warna HSI/HSV. Bandingkan hasilnya.
2. Lakukan noise reduction (noise removal) pada citra warna (citra peppers2.jpg) dengan mengolah masing-masing lapisan (R, G, dan B) kemudian menggabungkannya kembali. Pilih filter yang sesuai (derau pada citra adalah derau jenis salt & pepper).
3. Lakukan edge detection pada citra warna (citra peppers.png) dengan mengolah masing-masing lapisan (R, G, dan B) kemudian menggabungkannya kembali. Gunakan operator deteksi tepi yang pernah diberikan dalam pertemuan sebelumnya.



Thank You!